new model, an application can interact with the infrastructure and shape it the way required. It is not about designing the infrastructure with the application's requirement as the central theme (application-centric infrastructure); rather, it is about designing the infrastructure in a way that the application can drive it (application-driven infrastructure). We are not going to build infrastructure to host a group of applications but rather, we will create applications that can control various items of the infrastructure. Some of the prominent use cases involve applications being able to automatically recover from infrastructure failures. Also, scaling to achieve the best performance-to-cost ratio is achieved by embedding business logic in the application code that drives infrastructure consumption.

In today's competitive world, these benefits can provide a winning edge to a business against its competitors. While IT leaders such as Google, Amazon, Facebook and Apple are already operating in these ways, traditional enterprises are only starting to think and move into these areas. They are embarking on a journey to reach the ADDC nirvana state by taking small steps towards it. Each of these small steps is transforming the traditional enterprise data centres, block by block, to be more compatible for an application-driven data centre design.

Figure 1: Application-centric infrastructure

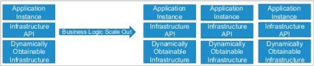

Figure 2: Application-driven infrastructure

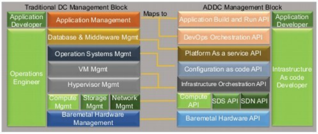

Figure 3: Traditional data centre mapped to ADDC

The building blocks of ADDC

For applications to be able to control anything, they require the data centre components to be available with an application programming interface (API). So the first thing enterprises need to do with their infrastructure is to convert every component's control interface into an API. Also, sometimes, traditional programming languages do not have the right structural support for controlling these infrastructure components and, hence, some new programming languages need to be used that have infrastructure domain-specific structural support. These languages should be able to understand the infrastructure components such as the CPU, disk, memory, file, package, service, etc. If we are tasked with transforming a traditional data centre into an ADDC, we have to first understand the building blocks of the latter, which we have to achieve, one by one. Let's take a look at how each traditional management building block of an enterprise data centre will map into an ADDC set-up.

1. The Bare-metal-as-a-Service API

The bare metal physical hardware has traditionally been managed by the vendor-specific firmware interfaces. Nowadays, open standard firmware interfaces have emerged, which allow one to write code in any of the application coding languages to interact through the HTTP REST API. One example of an open standard Bare-metal-as-a-Service API is Redfish. Most of the popular hardware vendors are now allowing their firmware to be controlled through Redfish API implementation. The Redfish specifications-compatible hardware can be directly controlled through a general application over HTTP, and without necessarily going through any operating system interpreted layer.

2. The software defined networking API

A traditional network layer uses specialised appliances such as switches, firewalls and load balancers. Such appliances have built-in control and data planes. Now, the network layer is transforming into a software defined solution, which separates the control plane from the data plane.

In software defined solutions for networking, there are mainly two approaches. The first one is called a software defined network (SDN). Here, the central software control layer installed on a computer will control several of the network's physical hardware components to provide the specific network functionality such as routing, firewall and load balancers. The second one is the virtual network function (VNF). Here, the approach is to replace hardware components on a real network with software solutions on the virtual network. The process of creating virtual network functions is called network function virtualisation (NFV). The software control layers are exposed as APIs, which can be used by the software/application codes. This provides the ability to control networking components from the application layer.

3. The software defined storage API

Traditional storages such as SAN and NAS have now transformed into software defined storage solutions, which can offer both block and file system capabilities. These software defined storage solutions are purpose-built operating systems that can make a standard physical server exhibit the properties of a storage device. We can format a standard x86 server with these specialised operating systems, to create a storage solution